

Caleb Bettcher

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EDUCATION

Cornell University

Aug 2026 – May 2027

Master of Computer Science

New York, NY

- Recipient of a Cornell Tech merit scholarship

University of Colorado Boulder

Aug 2023 – May 2026

Bachelor of Science in Computer Science, Minor in Business

Boulder, CO

- 3.92 Cumulative GPA, Dean's List, Summa Cum Laude, Accelerated 3-year track
CU Quants - Trading Team – CU Triathlon Team & CU Freeride - Member

WORK EXPERIENCE

Belvedere Trading

Summer 2026

Quantitative Software Engineering Intern

Boulder, CO

- Develop low-latency algorithms for making real time decisions in electronic trading systems
- Apply quantitative and statistical methods to identify market inefficiencies
- Interface with market data pipelines and APIs in performance-critical environments

Handshake AI Solutions

Aug 2025 – June 2026

Prompt Engineer / AI Research Fellow

- Design and implement structured, complex prompts in niche academic fields such as SAT/SMT solvers and advanced algorithmic reasoning to evaluate and fine-tune Large Language Models
- Analyze and resolve internal model reasoning failures to improve reliability and interpretability of AI systems

YouTube

Apr 2020 – Present

Content Creator

- Built a YouTube channel with over 80,000 subscribers and 10 Million views across 150 videos
- Hired an editor to streamline video production, and help generate revenue through sponsorships with companies such as Supercell

Net-Results Marketing Automation

Summer 2022 & 2023

Full Stack Web Development Intern

Denver, CO

- Built React frontend and GraphQL backend features enabling customizable automated marketing for 20,000+ active users

RESEARCH & PROJECTS

Quantitative Trader - CU Quants - University of Colorado Boulder

January 2026 – May 2026

- Monitored live market-making systems across 12 stablecoin instruments on Kraken US and OKX US, helping achieve a 98.44% uptime across a 1-178 bps spread capture range
- Made real-time decisions during periods of unexpected market movement, contributing to 137.63% annualized returns
- Contributed to proprietary trading dashboard and custom asset allocation engine, preparing systems for expansion to Gemini US

SurfaceEdge - Novel CNN Options Pricing Model from Surface Images

January 2026 – May 2026

- Engineered a novel multimodal deep learning pipeline combining CNN-encoded options surface images with contract-level scalars across 200M+ labeled contracts from 17 years of historical data, extending prior computer vision research from CU ML lab work
- Designed and compared three model architectures achieving 7.9% improvement in MAE over the naive baseline
- Adapted a causal self-attention transformer to model price history as a sequence prediction problem, finishing as our top model

Machine Learning Research - University of Colorado Boulder

Aug 2024 – May 2025

- Developed custom methodology software to generate datasets from multi-dimensional satellite data spanning over 6 years
- Trained and evaluated custom YOLO models in PyTorch to identify particle-precipitation signatures in optimized plots of satellite data
- Presented on the application and optimization of ML/AI to astrophysics at a national research conference

SKILLS & EXPERTISE

- **Languages** - Python, JavaScript, Julia, C++, Java, SQL
- **Frameworks/Libraries** - PyTorch, YOLO, React, Laravel, GraphQL
- **Technical Skills** - Git, HPC Clusters, Machine Learning, Computer Vision, Data Analysis, Financial Analysis
- **Professional Skills** - Critical Thinking, Project Management, Leadership, Communication, Technical Writing
- **Certifications** - TestOut - Network Pro, TestOut - Linux Pro, Handshake - Model Validation 2 (Expert)